

What advanced analytics does

Report View: Move beyond charts to explain patterns, drivers, and outliers



Choose the right advanced feature

Understand Drivers

Explain what influences outcomes

- ✓ Key Influencers for factor analysis
- ✓ Decomposition Tree for root cause
- ✓ AI splits to find highs and lows
- ✓ Smart Narrative for written summary

Detect and Explore

Find patterns and anomalies

- ✓ Anomaly Detection on line charts
- ✓ Natural language explanations
- ✓ Q&A for ad hoc exploration
- ~ Q&A retiring December 2026

Custom Visuals

Go beyond native Power BI charts

- ✓ R visuals for statistical plots
- ✓ Python visuals for ML outputs
- ✓ Heatmaps and distribution plots
- ~ Requires local engine setup

Key questions before choosing an advanced visual

What is influencing this outcome?

Why did this metric change?

Is this spike unusual?

Which segment is driving the result?

Do users need to ask questions freely?

Is a native visual enough here?

"A strong analyst does not just show what happened. They help users understand why it happened and where to investigate next."

What Are Advanced Visuals?

Advanced visuals in Power BI help users go beyond standard reporting. A normal report may show totals, trends, comparisons, and breakdowns. Advanced analytics visuals help users answer deeper questions such as:

- What is influencing this outcome?
- Why did this metric change?
- Which segment is driving the result?
- Is this spike unusual?
- Can the report summarize the insight for me?
- Do I need a custom statistical visual?

These features are useful when the report needs to become more than a collection of charts. They help turn Power BI into a more complete analytical experience where users can explore drivers, detect outliers, understand root causes, and generate insight from the dataset. **Advanced analytics helps Power BI move from reporting numbers to explaining patterns.**

Key Advanced Analytics Features in Power BI

The main features covered this week are:

- Key Influencers
- Decomposition Tree
- Smart Narrative
- Anomaly Detection
- Q&A
- R and Python Visuals

Each feature supports a different type of analysis. Some are useful for performance monitoring, some for driver analysis, some for natural language exploration, and some for advanced statistical visualisation.

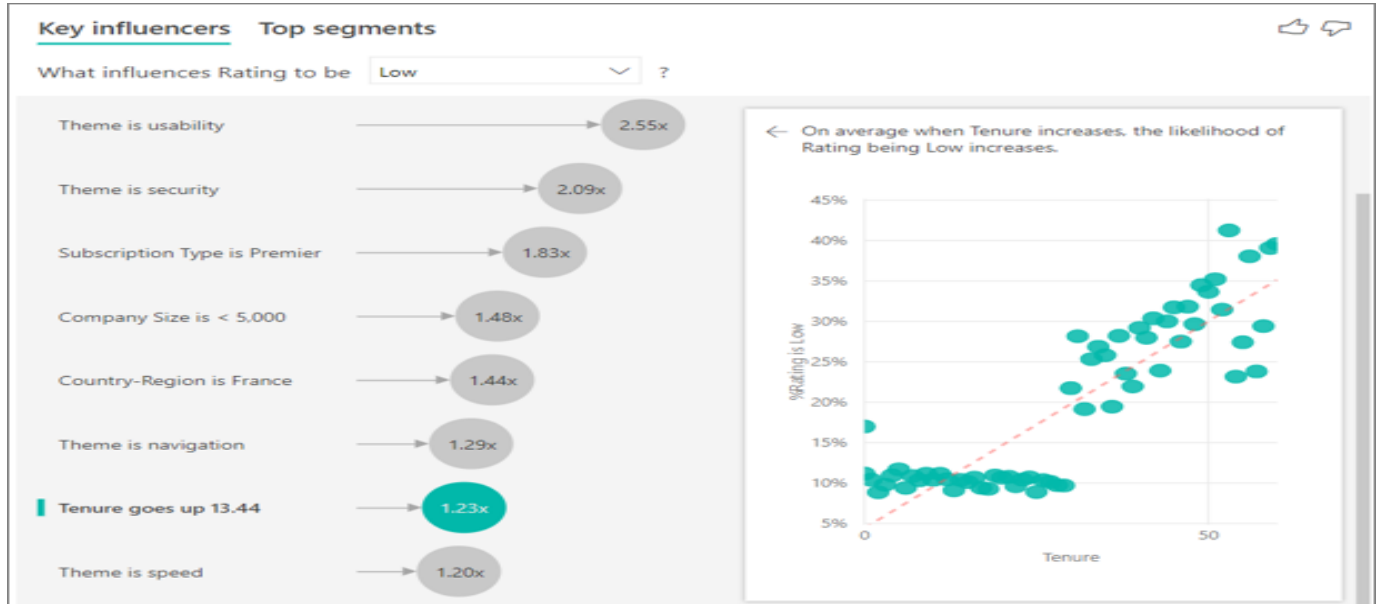
Key Influencers

The Key Influencers visual helps identify which factors are most likely to influence a selected outcome. For example, it can help answer:

- What influences customer churn?
- What drives low satisfaction?
- What factors increase the likelihood of high sales?
- Which attributes are linked to a selected outcome?

In Power BI, users place the outcome in the **Analyze** field and the possible explanatory fields in **Explain by**. The visual then shows which factors are associated with the selected result.

Use Key Influencers when: Users need to understand what factors may be driving a business outcome.



Decomposition Tree

The Decomposition Tree helps users break down a metric across multiple dimensions. It is useful for root-cause analysis because users can start with one number and drill down through categories such as region, product, segment, date, or customer type. For example:

Total Sales → Region → Product Category → Product → Customer Segment

Decomposition Tree is useful for visualizing data across multiple factors and drilling into the next dimension based on the analysis. Power BI also includes AI split options in the Decomposition Tree, allowing users to find high or low values automatically in the data.

Use Decomposition Tree when: Users need to break down a metric and investigate root causes.



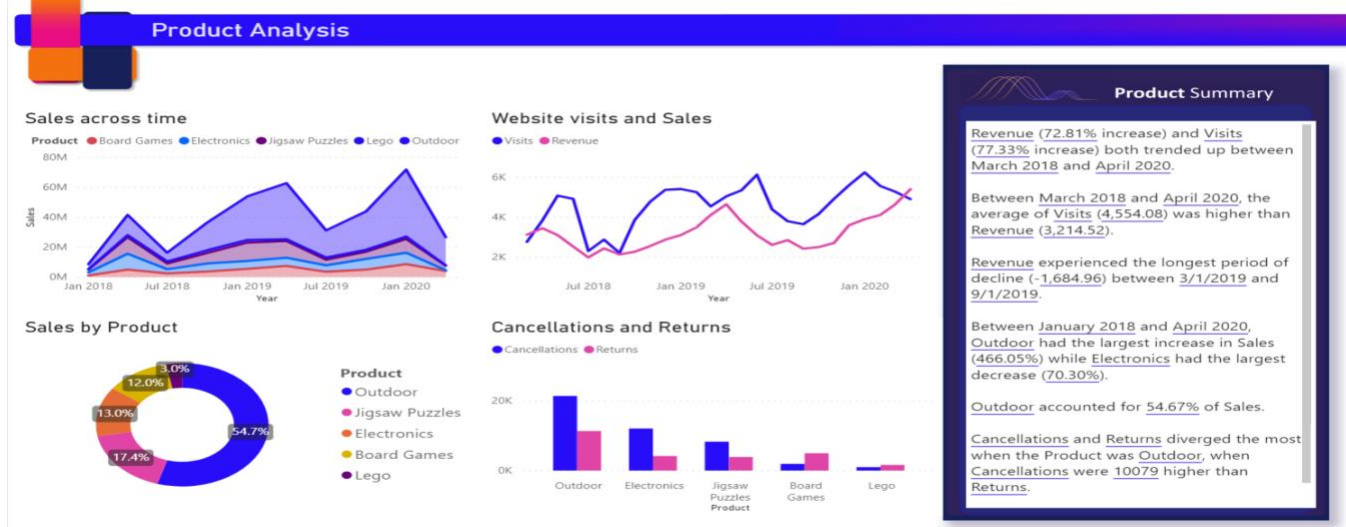
Smart Narrative

Smart Narrative creates written summaries of report visuals and data. It can help users understand key insights without manually writing every explanation. The text can update dynamically when users interact with the report through slicers or cross-filtering. For example, Smart Narrative can summarize:

- Which category had the highest sales
- Whether revenue increased or decreased
- Which segment contributed most to the total
- How a metric changed over time

Smart Narrative is a quick textual summary of visuals and reports that can identify useful insights and update dynamically through cross-filtering.

Use Smart Narrative when: Users need a written explanation of important visuals or report-level insights.



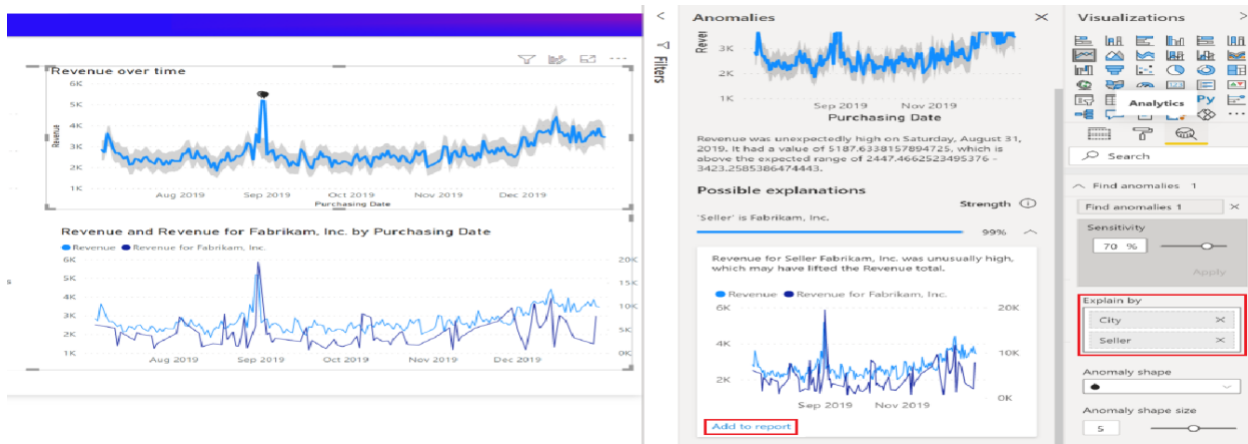
Anomaly Detection

Anomaly Detection helps identify unexpected spikes or drops in time-series data. It is useful when users need to monitor changes over time and quickly identify values that fall outside expected patterns. For example:

- Sudden increase in sales
- Unexpected drop in website traffic
- Unusual rise in returns
- Abnormal demand pattern
- Unexpected change in revenue

Power BI anomaly detection works with line charts and can generate explanations for detected anomalies. It uses AI-driven methods to highlight unexpected spikes or drops and can include natural language explanations for root-cause analysis. Anomaly Detection is available for line charts and helps find anomalies in time-series data.

Use Anomaly Detection when: Users need to identify unusual changes in time-based data.



Q&A Visual

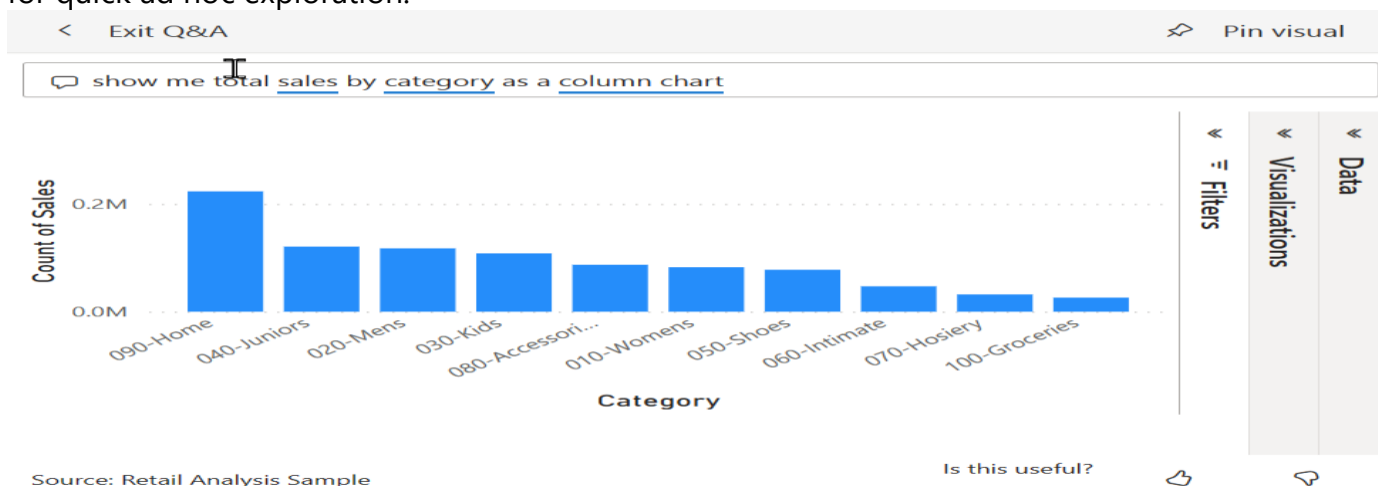
The Q&A visual allows users to ask questions about their data using natural language. For example, users can type:

- "total sales by region"
- "average profit by month"
- "top products by revenue"
- "sales trend over time"

Power BI then attempts to return an answer as a visual. Q&A relies on natural language processing, can be used in reports and dashboards, suggests starter questions, and can be converted into a standard visual.

Important note: Microsoft currently states that Power BI Q&A experiences are going away in December 2026 and recommends Copilot for Power BI as a more advanced replacement.

Use Q&A when: Users need to explore the dataset through natural language questions, especially for quick ad hoc exploration.



R and Python Visuals

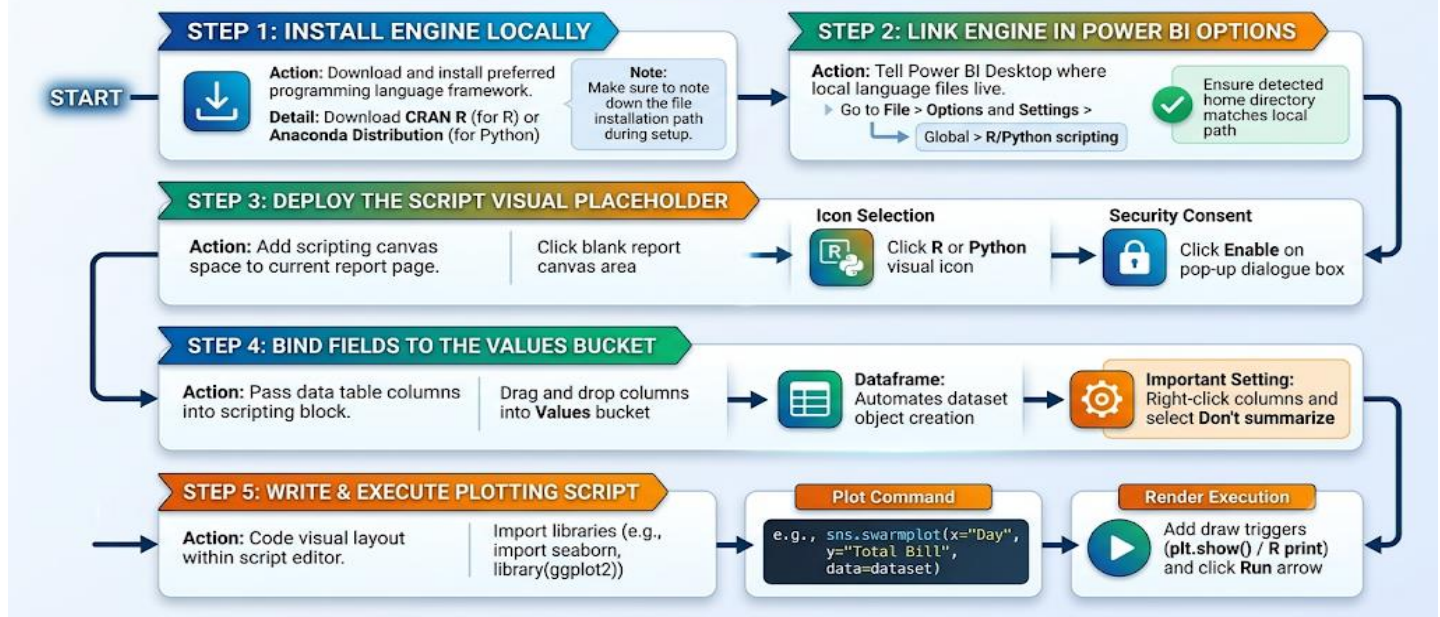
R and Python visuals allow analysts to create custom script-based visuals inside Power BI. They are useful when native Power BI visuals are not enough, especially for:

- Advanced statistics
- Machine learning outputs
- Custom plots
- Correlation analysis
- Heatmaps
- Distribution plots
- Specialized analytical visuals

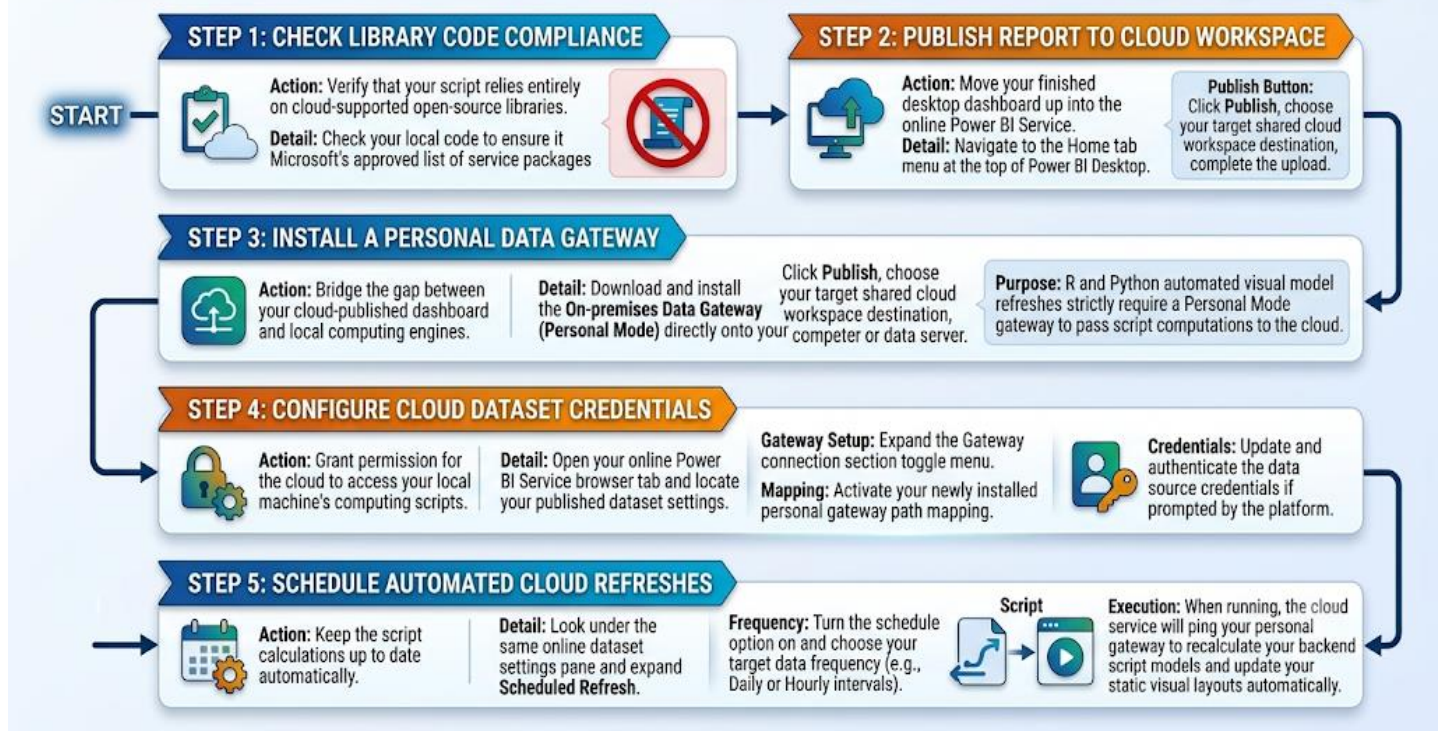
Native visuals are better when interactivity and cross-filtering are required, while R/Python visuals are stronger for complex statistics or machine learning.

Use R or Python visuals when: Users need advanced analysis or custom visuals that are not available through native Power BI charts.

ENABLING R AND PYTHON VISUALS IN POWER BI DESKTOP



PUBLISHING AND MAINTAINING SCRIPTS IN POWER BI SERVICE (CLOUD)



Practical Way to Decide



A Simple Decision Guide for Power BI AI & Advanced Visuals

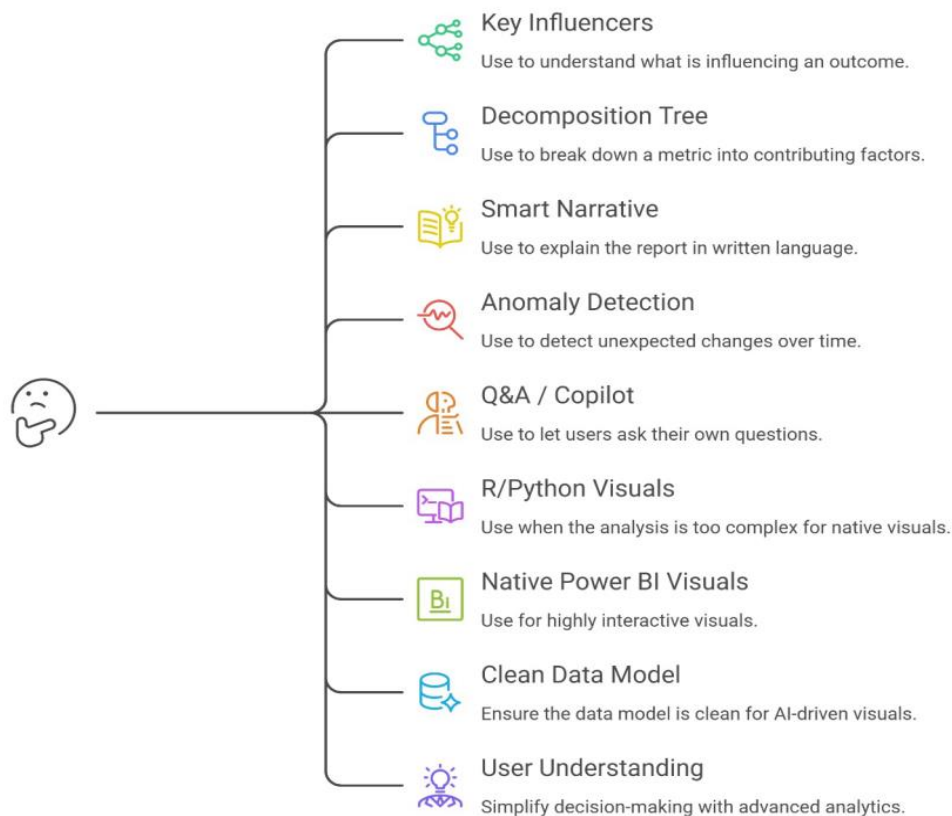
Choose the right advanced visual based on the insight or analysis question you are solving.

1 KEY INFLUENCERS		Use Key Influencers when: You need to understand which factors are influencing a selected outcome.
2 DECOMPOSITION TREE		Use Decomposition Tree when: You need to break down a metric across multiple dimensions and investigate root causes.
3 SMART NARRATIVE		Use Smart Narrative when: You need a dynamic written summary of visuals or report insights.
4 ANOMALY DETECTION		Use Anomaly Detection when: You need to identify unusual spikes or drops in time-series data.
5 Q&A		Use Q&A when: Users want to ask natural language questions and explore the dataset quickly.
6 R OR PYTHON VISUALS		Use R or Python Visuals when: You need advanced statistics, custom plots, or machine learning-based visuals.
7 NATIVE VISUALS		Use native Power BI visuals when: You need full report interactivity, cross-filtering, tooltips, and faster rendering.



The right advanced visual helps turn data into clearer insights and smarter decisions.

What Analysts Should Ask



Conclusion

Advanced analytics in Power BI helps reports move beyond basic charts and into deeper explanation. Key Influencers identify drivers, Decomposition Tree supports root-cause analysis, Smart Narrative summarizes insights, Anomaly Detection highlights unusual changes, Q&A enables natural language exploration, and R/Python visuals support custom analytical depth. The goal is not to use advanced visuals everywhere. The goal is to choose the right analytical feature for the question being asked.

A strong analyst does not just show what happened. They help users understand why it happened, what influenced it, and where to investigate next.

Stay tuned for next week's article in **The Insight Stack: Power BI, Layer by Layer**, where we move into **Week 10: Performance** and explore how analysts improve report speed, model efficiency, refresh behaviour, and user experience inside Power BI.

Further Reading:

- Decomposition Tree: [Create and View Decomposition Tree Visuals in Power BI - Power BI | Microsoft Learn](#)
- Key Influencers: [Key Influencers Visualizations Tutorial - Power BI | Microsoft Learn](#)
- Smart Narrative: [Create Smart Narrative Summaries - Power BI | Microsoft Learn](#)
- Anomaly Detection: [Anomaly Detection Tutorial for Power BI - Power BI | Microsoft Learn](#)
- Creating Power BI Visuals using R: [Create Power BI Visuals Using R - Power BI | Microsoft Learn](#)
- Creating Power BI Visuals using Python: [Create Power BI visuals using Python in Power BI Desktop - Power BI | Microsoft Learn](#)
- Analytics Pane: [Use the Analytics pane in Power BI - Power BI | Microsoft Learn](#)
- Exploring Power BI's Q&A feature: [Learn how to use natural language to explore data with Power BI Q&A - Power BI | Microsoft Learn](#)